

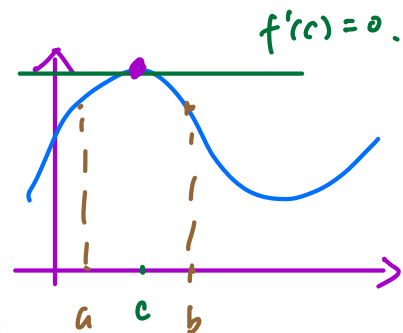
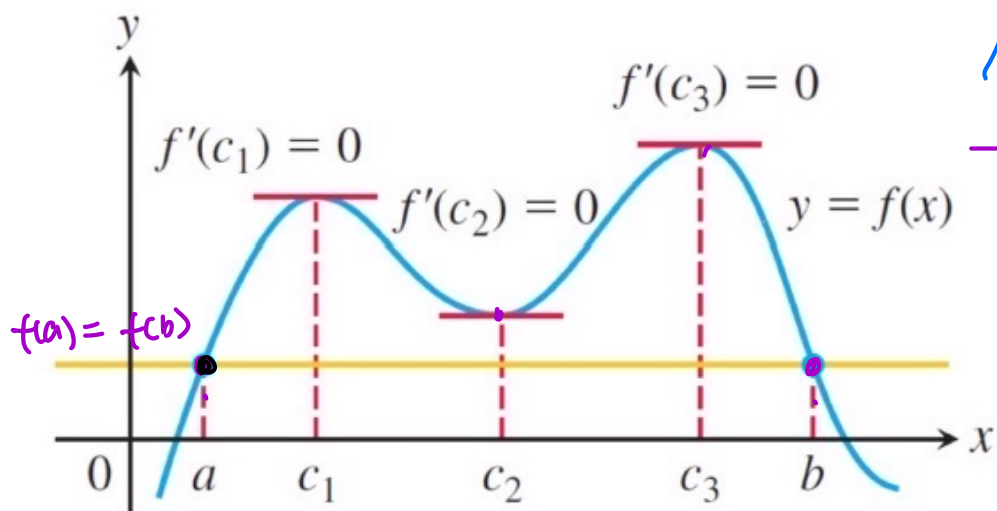
Lec 4-1. (Mon, Sep 22, 2025)

- Rolle's theorem
- Mean value theorem
- Monotonicity of functions
 - Intervals of monotonicity
 - First derivative test

· Rolle's theorem

Theorem [Rolle's theorem] Suppose that a function f is continuous on $[a, b]$ and differentiable on (a, b) , and it satisfies $f(a) = f(b)$.

Then there exists $c \in (a, b)$ such that $f'(c) = 0$.

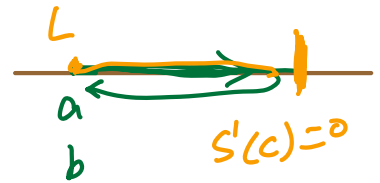


Rmk. A physical consequence of Rolle's thm is that:

if an object starts and stops a movement at the same position, and if the curve is differentiable, then at some point in between, it must stop moving.

$$s(a) = s(b) = L.$$

Pf of Rolle's thm: [Idea: they are local extrema!]



Step 1. By Extreme Value Theorem, $\exists c_1, c_2 \in [a, b]$.

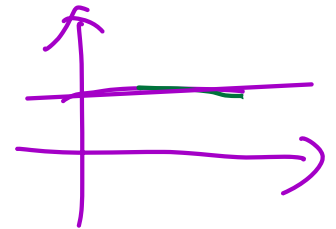
$$\text{s.t. } f(c_1) = m, f(c_2) = M,$$

where m and M are absolute min and max respectively.

They are local extrema as well.

Step 2. If $m = M$, then

$$f(x) \equiv \text{const } (=m=M), \forall x \in [a, b].$$



$$\text{Then } f'(x) \equiv 0 \quad \forall x \in (a, b).$$

We can choose any $c \in (a, b)$ such that $f'(c) = 0$ (e.g., $c = \frac{a+b}{2}$).

Step 3. If $m < M$, then

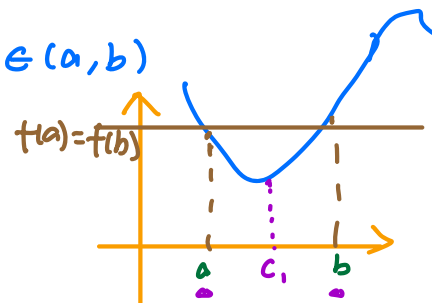
either $f(a) = f(b) \neq m$ or $f(a) = f(b) \neq M$ or both hold.

We suppose $f(a) = f(b) \neq m = f(c_1)$. We analyse the local min point.

Then ① c_1 is not endpoint as $c \neq a, b$ and hence $c \in (a, b)$

② f is differentiable at c by conditions.

$\therefore$ By Thm 4.1.2, $f'(c_1) = 0$.



Step 4. If $f(a) = f(b) \neq M = f(a)$ similar: we take $c = c_2$. (local max) □.

e.g.5. Using ^{all} calculus theory above, prove that if $a > 0$ and $b > 0$, then $ax^3 + bx + d = 0$ has exactly one solution.

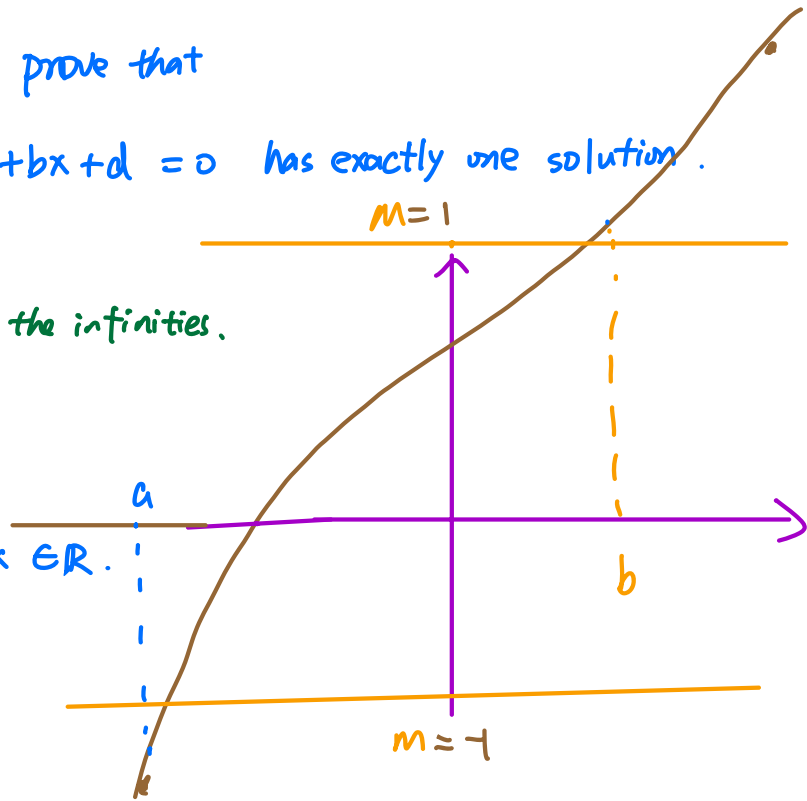
Pf: Step 1: analyze the derivative and the infinities.

Write $f(x) \triangleq ax^3 + bx + d$.

$$\therefore f'(x) = 3ax^2 + b > 0, \quad \forall x \in \mathbb{R}.$$

$$\text{And } \lim_{x \rightarrow -\infty} f(x) = -\infty,$$

$$\lim_{x \rightarrow \infty} f(x) = \infty$$



Step 2. A bounded interval. $[a, b]$

By definition, for any $M > 0$, there exists some $\sim b \in \mathbb{R}$, such that $f(x) > M$, for any $x \geq b$.

For any $m < 0$, there exists some $\sim a \in \mathbb{R}$, such that $f(x) < m$, for any $x \leq a$.

Let $M = 1$ and $m = -1$. We find a and b above,

s.t. $f(a) < m = -1$, $f(b) > M = 1$.

Step 3: Intermediate value thm (IVT): $[a, b]$

We use IVT for $0 \in [f(a), f(b)]$ (i.e., $f(a) < -1 < 0 < 1 < f(b)$).

As f is continuous on $[a, b]$, we can find some $c \in [a, b]$

such that $f(c) = 0$.

$\therefore c$ is the solution of $f(x) = 0$.

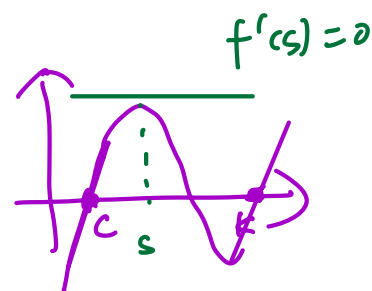
Step 4. Uniqueness (Rolle's Thm). $[a, b]$

If there exists another solution $k \neq c$, s.t. $f(k) = 0$.

Then either $k > c$ or $k < c$.

Suppose $k > c$.

Then $f(k) = f(c) = 0$.



As f is diff' on (c, k) and cont' on $[c, k]$,

we can find $s \in (c, k)$, s.t. $f'(s) = 0$.

But $f'(x) > 0$, $\forall x \in \mathbb{R}$.

Then we have a contradiction! Hence, solution c is unique. $\square$

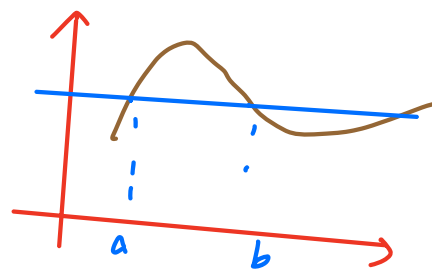
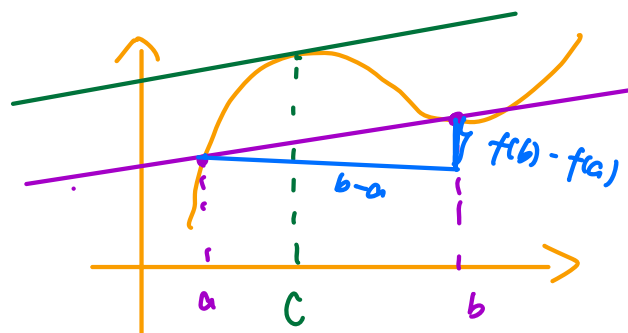
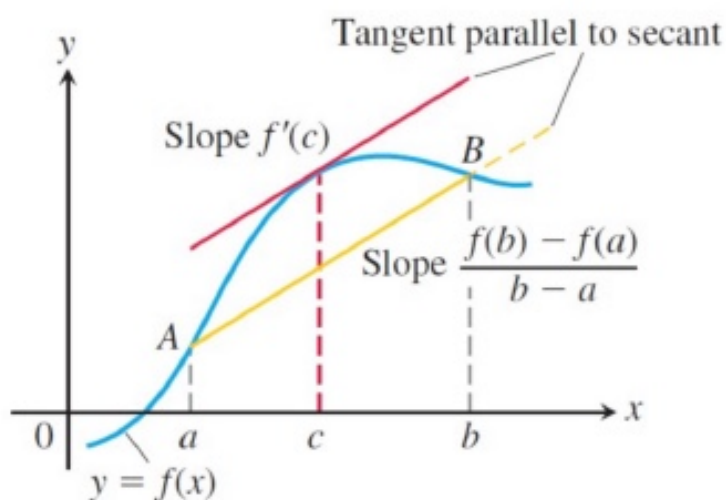
• Mean Value theorem

One of the most fundamental and important thms in calculus!

Thm (Lagrange's Mean Value Theorem).

Suppose that a function f is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Pf : [Idea : use Rolle's theorem] .

[Construct a function such that

$$\cdot g(a) = g(b)$$

$$\cdot g'(x) \Big|_{x=c} = \quad \quad \quad = 0 \quad]$$

Write $g(x) = f(x) - \frac{f(b) - f(a)}{b - a} (x - a) \quad , \quad x \in [a, b]$.

Check : $\cdot g(x)$ is cont' on $[a, b]$ and diff' on (a, b)

$$\cdot \underline{g(a)} = f(a) - \frac{f(b) - f(a)}{b - a} \cdot 0 = f(a) , \quad \underline{g(b)} = f(b) - \frac{f(b) - f(a)}{b - a} \cancel{(b - a)} = f(a)$$

$\therefore$ By Rolle's thm, there exists $c \in (a, b)$ such that

$$g'(c) = 0 \quad \Rightarrow \quad f'(c) - \frac{f(b) - f(a)}{b - a} = 0$$

$$\Rightarrow f'(c) = \frac{f(b) - f(a)}{b - a} .$$

□ .

A Physical Consequence of the MVT

Suppose that $f(t)$ represents the distance travelled until time t . Then the mean value theorem implies that if we pick two moments $t = a$ and $t = b$, there has to be some moment $t = c$ in between at which the instantaneous speed is equal to the average speed between $t = a$ and $t = b$.



→ <https://mrhonner.com/archives/10268>

Mr Honner

ABOUT

Calculus Gave Me a Speeding Ticket

Published by [MrHonner](#) on April 9, 2013

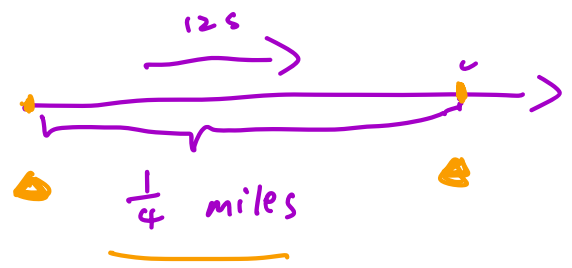
Years ago, one sunny Sunday afternoon, I was driving home from visiting friends at college and received a speeding ticket.

I didn't realize it at the time, but calculus played an important role in my citation.

You see, this was no ordinary speeding ticket, the kind where a police officer paces the offender or uses radar to measure a vehicle's speed. My speed was calculated from an airplane high above the road. And the Mean Value Theorem clinched the case.



Aerial speed enforcement works like this: large marks painted on the road divide the highway into quarter-mile intervals. A pilot flying overhead uses a stopwatch to time a suspected speeder from one mark to the next. Say the pilot records a time of 12 seconds; a simple calculation converts one quarter mile per 12 seconds into 75 miles per hour; this information, the *average speed on this interval*, is radioed to the police on the ground who then stop and ticket the driver.



• distance: $\frac{1}{4}$ miles ✓

• measure: 12 s ✓ $\Rightarrow$ average speed = $\frac{(\frac{1}{4} - 0) \text{ miles}}{12 \text{ s}}$

$$= \frac{\frac{1}{4} \times 300 \text{ miles}}{3600 \text{ s}}$$

$$= 75 \text{ miles/hour}$$

$$\approx 75 \times 1.6 \text{ km/h}$$

$$= \underline{120} \text{ km/h}$$

$$> 100 \text{ km/h}$$

• MVT $\Rightarrow \exists t_0 \in (0, 12s),$
 s.t. $f'(t_0) = 120 \text{ km/h}$
 $> 100 \text{ km/h}$

[Already over the limit!]

[For some other points, the speed is even higher!]

• Math consequences of MVT.

Corollary 1. If $f'(x) \equiv 0$ at each point x of an open interval I , then $f(x) \equiv \text{const}$ for all $x \in I$, where C is a constant.

Pf: We prove by contradiction.

Suppose we have $x_1, x_2 \in I$ and $x_1 \neq x_2$ such that

$$f(x_1) \neq f(x_2) .$$

Without loss of generality, we assume $x_1 < x_2$.

As f is continuous on $[x_1, x_2]$ and differentiable on (x_1, x_2) ,

by MVT, we have there exists some $c \in (x_1, x_2)$ s.t.

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \neq 0 .$$

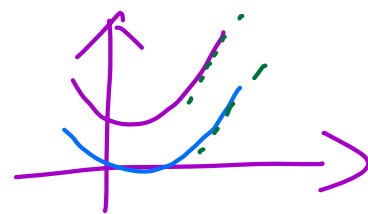
which is a contradiction to the condition $f'(x) \equiv 0, \forall x \in I$.

$\therefore f(x) \equiv \text{const} , \quad \forall x \in I .$ □

Corollary 2. If $f'(x) = g'(x)$ at each point x in an open interval I ,

then there exists a constant C such that

$$f(x) = g(x) + C, \quad \forall x \in I.$$



That is, $f - g$ is a constant function on I .

Pf: Let $h(x) \triangleq f(x) - g(x)$ and apply Corollary 1 to h . □

$$\therefore [h' = 0 \Rightarrow h = \text{const}] .$$

Eg. 1 If $f'(x) = \sin x$, what is $f(x)$?

Sol': Write $g(x) = -\cos x$. Then $g'(x) = \sin x = f'(x)$ for any $x \in \mathbb{R}$.

$\therefore f(x) = g(x) + C$ for some $C \in \mathbb{R}$, based on Cor 2.

$\therefore f(x) = -\cos x + C$ for some $c \in \mathbb{R}$. □

Eg. 2. If $f'(x) = 5x^2 - \frac{1}{\sqrt{x}}$ and $f(1) = 0$, what is $f(x)$?

Sol': Write $g(x) = \frac{5}{3}x^3 - 2x^{\frac{1}{2}}$.

Then we find $g'(x) = 5x^2 - \frac{1}{\sqrt{x}} = f'(x)$!

By Cor 2, $f(x) = g(x) + C$ for some $C \in \mathbb{R}$

$\therefore C = f(1) - g(1) = 0 - \left(\frac{5}{3}x^3 - 2x^{\frac{1}{2}}\right)\bigg|_{x=1} = -\left(\frac{5}{3} - 2\right) = \frac{1}{3}$

$$\therefore f(x) = g(x) + \frac{1}{3} = \frac{5}{3}x^3 - 2x^{\frac{1}{2}} + \frac{1}{3}.$$

□.

check: $f'(x) = 5x^2 - x^{-\frac{1}{2}} + 0.$ ✓

$$f(1) = \frac{5}{3} - 2 + \frac{1}{3} = 0.$$

The MVT can be used to prove inequalities.

e.g.3. Prove that:

$$(a) |\sin y - \sin x| \leq |y - x| \text{ for all } x, y \in \mathbb{R}.$$

Pf: If $y = x$, the inequality holds trivially [both sides = 0].

Next we assume $y > x$ without loss of generality.

As the function $f(z) = \sin z$ is differentiable ^{and} (hence continuous) on $\mathbb{R}$,

by MVT, there exists some $c \in (x, y)$, s.t.

$$\cos(c) = f'(c) = \frac{f(y) - f(x)}{y - x} = \frac{\sin y - \sin x}{y - x}.$$

$$\Rightarrow \cos(c) = \frac{\sin y - \sin x}{y - x}$$

$$\therefore \sin y - \sin x = \cos(c) \cdot (y - x)$$

$$\therefore |\sin y - \sin x| = |\cos c| |y - x| \leq 1 \cdot |y - x|.$$

□

$$(b) \quad \frac{1}{2\sqrt{x_0+1}} < \sqrt{x_0+1} - \sqrt{x_0} < \frac{1}{2\sqrt{x_0}} \quad \text{for all } x_0 > 0.$$

Pf : Write $f(x) = \sqrt{x}$. Fix $x_0 > 0$.

As $f(x)$ is differentiable (and hence cont') on $(0, \infty)$,

by MVT, there exists some $c \in (x_0, x_0+1)$, s.t

$$f'(c) = \frac{f(x_0+1) - f(x_0)}{x_0+1 - x_0} .$$

$$\Rightarrow \quad \frac{1}{2} c^{-\frac{1}{2}} = \frac{(x_0+1)^{\frac{1}{2}} - x_0^{\frac{1}{2}}}{1} .$$

$$\therefore \quad x_0 < c < x_0 + 1$$

$$\therefore \quad \frac{1}{2} (x_0+1)^{-\frac{1}{2}} < \frac{1}{2} c^{-\frac{1}{2}} < \frac{1}{2} x_0^{-\frac{1}{2}}$$

$$\therefore \quad \frac{1}{2} \frac{1}{\sqrt{x_0+1}} < (x_0+1)^{\frac{1}{2}} - x_0^{\frac{1}{2}} < \frac{1}{2} \frac{1}{\sqrt{x_0}} .$$

$$\therefore \quad \frac{1}{2\sqrt{x+1}} < \sqrt{x+1} - \sqrt{x} < \frac{1}{2\sqrt{x}} \quad \text{for any } x > 0. \quad \square .$$